Neighbourhood-aware misbehavior detection for V2X

Research plan: problem, hypotheses, experimental design and validation. Status: final. Main study on 14 of 19 attack types, collusion study done, v2 detector done, paper drafted (paper/main.tex); extension to 18 of 19 attack types (adds the 4 Sybil attacks) done. Results quoted are final (14 attacks: mean of 2 seeds; 18 attacks: 3-model ensemble).

1. Problem formulation

Specific ML/AI problem or failure mode addressed

Connected vehicles broadcast Basic Safety Messages (position, speed, heading). A misbehaving vehicle can send false data, for example a random or spoofed position or a frozen speed, which misleads other vehicles. Most learned detectors judge each sender **in isolation**, so a sender's lie is only visible against its own history. They ignore what the surrounding vehicles report, and the evidence from neighbours that could expose an inconsistency goes unused.

The failure mode we address has two parts:

- **Missed or slow detection of falsified data** when a sender is judged alone (lower recall, longer time to first alarm).
- **Unknown behaviour when neighbours are themselves malicious.** Cooperative schemes often assume an honest majority and fail under collusion (2025 V2X intrusion-detection survey); it is untested whether a neighbour-aware learned detector degrades when many neighbours are attackers.

2. Hypothesis formulation

Primary hypothesis

Null hypothesis (H0): A detector that also attends to the 16 nearest neighbours has the same detection quality (F1) as the same model without neighbour attention on held-out traffic: $F1_{full} = F1_{no\text{-}nbr}$.

Alternative hypothesis (H1): The neighbourhood-aware detector achieves higher F1 than the same model without neighbour attention: $F1_{full} > F1_{no\text{-}nbr}$, without raising the false-positive rate.

Secondary hypothesis (robustness to malicious neighbours)

H0b: Recall of the neighbour-aware detector falls as the share of malicious neighbours rises (it is poisoned by them).

H1b: Recall stays within a small margin (e.g. 2 points) of its value at 0 to 10% malicious neighbours, even above 50%.

3. Research and experimental design

Item	Choice
Dataset	VeReMi Extension (public benchmark, simulated traffic and attackers; F2MD / VEINS / SUMO, Luxembourg LuST scenario, 30% attackers). Zenodo record 20090854. Done: 14 of 19 attack types (14:00-16:00) for the main study, then 18 of 19 with the four Sybil attacks added (GridSybil, DataReplaySybil, DoSRandomSybil, DoSDisruptiveSybil); the RandomSpeedOffset archive was excluded as mislabeled. Not done: dense (07:00-09:00) traffic for v2, original VeReMi / VeReMi NextGen as external checks.

Item	Choice
Input / windows	Receiver-centric windows: K = 16 nearest senders x T = 10 one-second steps. v2 inputs per sender and step: 9 claim values + 10 plausibility checks (kinematics, road map learned from honest traffic, nearest other claim, exact self-repeat, copied trail, log movement). Split: 109,823 train, 29,547 validation, 68,971 test windows. **18-attack extension:** senders are tracked by pseudonym, which is all a receiver sees, so each fake Sybil identity is a separate sender; labels come from the true vehicle. 3 new inputs (22 in total): identity age (following pseudonym changes), handover distance, and lockstep (how rigidly a sender's offset to its most co-moving other sender stays fixed). 141,260 train windows (+51,033 augmented), 88,695 test windows.
Baseline	**Per-sender** model: the same transformer without neighbour information. Also the **no-spatial** ablation (sees neighbours, no cross-sender attention). Rule-based reference: range and speed-consistency checks from the original VeReMi paper. Also the first model (same design, no checks).
Proposed	**Final model (v2):** small factorised transformer (144k parameters) with temporal and cross-sender (neighbour) attention over the 19 inputs; training adds careful injected lies (offset, fake stop); a receiver-side memory carries evidence across windows. **18-attack system:** the mean score of 3 models (two 144k-parameter models, seeds 0 and 1, plus one wider and deeper model: width 96, 3 blocks, 16 epochs), threshold picked on validation.
Metrics	Precision, recall, F1, false-positive rate (FPR), AUC-ROC (per attack and overall), **detection latency** (seconds from a sender's first appearance to the first alarm), recall / FPR stratified by the share of malicious neighbours, and colluder recall at matched 1% FPR. Scored with message labels (attacker counted only in windows where it sent a falsified message) and vehicle labels (every window of an attacker vehicle).

4. Proposed validation strategy

Done so far

- Train on the first hour of the simulation, test on the second hour; validation and test are separate blocks of receivers, and thresholds and memory decay are chosen on validation only.
- 2 random seeds per model; results are the mean of the two.
- **Statistical significance:** paired bootstrap over receivers (2,000 resamples) with Holm correction across attacks.
- **Robustness:** results by malicious-neighbour share; a collusion stress test (coordinated consistent lies, colluder recall at matched 1% FPR).
- **Sybil attacks (done):** 18 of 19 attack types, scored with message and vehicle labels, with a no-neighbour ablation of the same ensemble.

Remaining / next steps

- **Missed GridSybil identities** look honest on every content check (on the road, consistent motion), and no timing or formation cue separated them; like on-road consistent lies, they need an independent signal.
- **External check** on original VeReMi and VeReMi NextGen.
- **Dense (rush-hour) traffic** for the v2 detector.
- **A 3rd seed** for every model.

- **On-road consistent lies** need an independent signal (signal strength, on-board sensors or roadside units); the claims alone are not enough.

5. Outcome (final results)

- **Detection quality (v2, message labels):** precision 98.65, recall 97.12, F1 97.88, accuracy 98.69, FPR 0.60%. With receiver memory: precision 98.16, recall 97.78, F1 97.97, accuracy 98.74.
- **Vehicle labels:** v2 precision 98.65, recall 95.41, F1 97.01, accuracy 98.14; with memory recall 96.09, F1 97.12, accuracy 98.20. First model (no checks): precision 98.4, recall 91.6, F1 94.9, accuracy 96.9, FPR 0.7%. Speed-check rule: F1 78.9 (vehicle) / 80.0 (message).
- **Neighbour attention (H1):** +0.3 F1 (95% CI about [0.2, 0.4]), significant on 8-10 of 14 attacks; FPR 0.90% -> 0.60%. H0 is rejected overall, but the gain is small.
- **Malicious neighbours (H1b):** the benefit shrinks and turns negative when attackers are the majority of neighbours (v2: -0.004 within-attack; first model: +0.071 -> -0.023). So H1b is not supported.
- **Latency:** mean 0.9 s to first alarm, median 0; 96% of attacker-receiver pairs flagged.
- **Collusion:** the first model caught almost none of the consistent lies (colluder recall at most 0.04). v2: 50 m offset 0.83-0.86, fake stop about 0.70, spoofed convoy never seen in training 0.87-0.90. Map-aware on-road lies are still hard: offset along the road 0.23-0.25, convoy on its own road 0.36-0.48. Coordinated convoys raised the first model's honest FPR from 0.9% to 3.0%.
- **Reference paper** (Abdel Hakeem & Kim, IEEE T-ITS 2025 survey, Table XIII: 97.37% on the original 5-attack VeReMi): v2 + memory is above 97.37 on all four metrics under message labels; under vehicle labels recall and F1 are not. This is not a like-for-like comparison.
- **18-attack extension (Sybil added; 88,695 test windows):** message labels precision 96.30, recall 95.46, F1 95.88, accuracy 97.04, FPR 2.07%; vehicle labels precision 96.33, recall 94.44, F1 95.38, accuracy 96.66. Without neighbour attention the same ensemble scores F1 94.45 (message) / 93.94 (vehicle), so neighbour attention adds about 1.4 F1 with Sybil present (vs +0.3 on 14 attacks).
- **Per attack (18-attack system, message labels, recall):** DoSRandomSybil 0.997, DoSDisruptiveSybil 0.969, DataReplaySybil 0.914, GridSybil 0.823 (the largest attack, 20,598 test slots); the 14 original attacks are mostly at or above 0.96 (DelayedMessages 0.912). On the 14 original attack runs this model scores message F1 about 97.3 with FPR about 1.6% (vs 0.6% for the 14-attack model), because it lowers its threshold to catch Sybil identities.
- **Tried and did not help:** a bigger model alone (message F1 95.19 vs 95.76 for the 2-seed small ensemble) and receiver memory (Sybil identities change pseudonyms).
- **Against the 97.37% reference:** the 14-attack system beats it on all four metrics (message labels + memory). On 18 attacks it does not: precision 96.3, recall 95.5, F1 95.9 and accuracy 97.0 are all below 97.37. The reference is the original 5-attack VeReMi with no Sybil attacks.
- **Paper:** drafted as paper/main.tex -> paper/main.pdf (IEEE conference format, 6 pages, 18 references).

Scope: the benchmark is simulated (VeReMi Extension), not real-world data. 18 of 19 attack types have been run (Sybil included; RandomSpeedOffset excluded as mislabeled); collusion is covered only by the stress test, and crash intent is not modelled.